Vaishnavi Khindkar

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Education

Carnegie Mellon University, School of Computer Science (Robotics Institute) *Aug 2025 - Dec 2026*
Master of Science in Computer Vision GPA 4.11/4.0

- Relevant Coursework: Advanced Computer Vision (16-820), Learning for 3D Vision (16-825), Robot Learning (16-831), Robot Localization & Mapping (16-833), Visual Learning & Recognition (16-824), Deep Learning Systems (10-714).

Technical Skills

Programming: Python (3.x), Modern C++ (14, 17, 20), MATLAB, SQL, C, R, Octave, Core Java
ML / CV: Machine Learning, Deep Learning, Computer Vision, 3D Vision, Multimodal Learning, Object Detection
Frameworks and Tools: PyTorch, TensorFlow, AWS, NVIDIA Triton, Linux, Git, LaTeX, Jupyter Notebook, OpenCV, Docker
Domains: Generative Modeling & Sampling, 3D & 4D Scene Understanding, Video Generative AI, Object-Centric & Spatial Representations, World Models, Autonomous Driving, Robotics

Experience

Sync Labs San Francisco Bay Area, CA
Machine Learning Engineering Intern *May 2026 - Present*

- Architected GPU-served VLM inference on Modal for automated visual quality control of AI lip-sync video, replacing manual, frame-by-frame review: deployed multimodal models (Google DeepMind's Gemma-3-12B, Qwen2.5-VL-7B), benchmarked against NVIDIA Cosmos Reason, and engineered structured-output prompting, multi-model agreement gating, and per-model routing around each model's blind spots.
- Built a human-in-the-loop labeling loop feeding a LoRA fine-tuning path for Qwen2.5-VL on artifact crops, and evaluated RLHF vs. supervised fine-tuning with a per-class precision/recall/F1 harness to drive model iteration.

Carnegie Mellon University, Machine Learning Department Pittsburgh, PA
Graduate Research Fellow (Advisor: Prof. Andrej Risteski) *May 2026 - Present*

- Studying reward-tilted sampling: principled lift-and-tilt methods for steering generative models toward target distributions under complex, indefinite rewards.

Carnegie Mellon University, Robotics Institute Pittsburgh, PA
Research Assistant (Advisor: Prof. Fernando De la Torre) *Aug 2025 - Present*

- Developed the first framework using foundation generative models (Imagen-3) for systematic diagnosis of aerial object detectors, enabling fine-grained failure analysis and up to 11% AP50 gains.

Product Labs (Bhashini, Government of India) Hyderabad, India
Applied AI Researcher *Oct 2023 - Jul 2025*

- Containerized OCR models with Docker and deployed via NVIDIA Triton, achieving real-time latency (0.11s printed, 0.051s handwritten) across 22 Indian languages.

CVIT, IIIT Hyderabad Hyderabad, India
Jr. Research Scientist (Advisors: Prof. C. V. Jawahar, Prof. Vineeth N. Balasubramanian, Prof. Chetan Arora) *Aug 2020 - Oct 2023*

- Led research on reason-enriched pedestrian intent estimation (MindReaD, IROS 2024), creating the PIE++ dataset and achieving +5.6% accuracy and +7% F1 over baselines.
- Developed ILLUME, a self-attention-based adaptive object detection framework (WACV 2022) with improved structural attention and state-of-the-art domain adaptation performance.

Barclays Pune, India
Software Developer *Aug 2018 - Oct 2020*

- Built a Contacts application with Spring Boot and AngularJS, improving UX and reducing system errors.

Select Publications and Patents

- Diagnosing Aerial-View Object Detectors with Foundational Image Generative Models *[Link]*
  Panev S., Jeon M., Khindkar V., Deshpande A., de Melo C.M., Hu S., Chakraborty S., De la Torre F. **(ECCV 2026)**
- Can Reasons Help Improve Pedestrian Intent Estimation? A Cross-Modal Approach *[Link]*
  Khindkar V., Balasubramanian V.N., Arora C., Subramanian A., Jawahar C.V. **(IROS 2024)**
- To miss-attend is to misalign! Residual Self-Attentive Feature Alignment for Adapting Object Detectors *[Link]*
  Khindkar V., Balasubramanian V.N., Arora C., Saluja R., Subramanian A., Jawahar C.V. **(WACV 2022)**
- GAMMA: Generative Augmentation for Attentive Marine Debris Detection *[Link]*
  Khindkar V., Khindkar J. **(arXiv 2022)**
- System and method for detecting object in an adaptive environment using a machine learning model
  Khindkar V., Balasubramanian V.N., Arora C., Subramanian A., Jawahar C.V. **(US Patent 2023)**

Select Projects

- **Slot4D: Learning Object Slots for 4D Reconstruction** - *CMU Capstone (Advisor: Prof. Laszlo A. Jeni)*
  Built an adaptive slot-attention head over VGGT features that decomposes scenes into object slots and masks; slot-conditioned refinement achieves up to 4× better surface reconstruction on CO3D, ScanNet, and Virtual KITTI; extending to streaming 4D.
- **Cross-Resolution Gaussian Splatting for Hybrid Resolution Photography** - *(Advisor: Prof. Shubham Tulsiani) [Link]*
  Designed a cross-resolution 3D Gaussian Splatting pipeline that fuses heterogeneous image sources, incorporating detail-preserving densification for high-fidelity reconstruction.
- **Generalizable Single-View Holistic Robot Pose Estimation** - *(Advisor: Prof. Jun-Yan Zhu)*
  Designed a single-view robot pose and keypoint estimation pipeline with synthetic-to-real transfer and domain randomization for unseen robot types.

Academic Service & Volunteering

Reviewer: CVPR 2025, ECCV 2024, CVPR 2023, ECCV 2022, WACV 2022 **Mentor:** AI4ALL Changemakers in AI (2022)
Teaching: CV tutorials for IISc-TalentSprint Exec. M.Tech **Organizer:** Hour of Code (Intl. Coding Week)